25900398 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCFTQ200 - FOOD PRESERVATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1. How does food preservation prevent microbial growth? [U] / [CO1]
2. What are the different types of refrigerant? [K] / [CO2]
3. What is the role of foam in foam mat drying? [U] / [CO3]
4. Write in short on water thawing [K] / [CO3, CO4]
5. Develop a note on hypertonic solution used in osmotic dehydration. [A] / [CO2, CO5]
6. What is the impact of moisture control on food shelf life? [K] / [CO1]
7. What is syruping and brining? How is it done? [U] / [CO2, CO4]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. List out the advantages & disadvantages of osmotic dehydration. [K] / [CO2]

9. Write short notes on (1) Plate freezing (2) Immersion freezing [K] / [CO3]

10. Write a brief note on radappertization, radurization and radicidation. [U] / [CO2]

11. Enumerate the preservation methods used in the killing of microorganisms. [K] / [CO1]

12. Explain in detail on flash 18 process. [U] / [CO2]

13. Explain the different types of hurdles used in combined preservation of foods. [K] / [CO2]

14. What do you mean by exhausting of cans? List the methods and advantages of exhausting. [K] / [CO2, CO4]

[4x5 = 20]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

15. Write an essay on the following types of pasteurization: (1) LTLT (2) HTST and (3) UHT. [U] / [CO2]

16. Define irradiation. Give an account of the major applications of irradiation in food industry. [A] / [CO2]

17. Write an essay on the types of sterilizers used in food industries. [U] / [CO2]

18. State the principle of hurdle technology. Discuss the application of hurdle technology in (1) Moisture foods (2) Heated foods (3) Chilled foods (4) Fermented foods. [K] / [CO2]

[10x2 = 20]